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The effect of institutional cross-ownership on corporate risk-taking in a transitional economy

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ABSTRACT

This study documents the unintended effect of institutional cross-ownership on corporate risk-taking in a transitional economy. Theoretically, cross-owners can increase their influence in corporate governance through voice and exit. However, such an insight may not align with the reality of the Chinese market, where institutional activism is limited. We predict and document that the amplified exit threat of institutional cross-ownership increases its contribution to managerial myopia and reduces risk-taking activities. This effect is more pronounced for firms when the exit threat is more credible or when managers are more concerned about market performance. As expected, the association is driven primarily by active short-term investors, whose exit threat is much stronger. Our findings suggest that institutional cross-ownership may inadvertently exacerbate agency problems because, in some cases, it may discourage managers from taking value-maximizing actions.

1. Introduction

Previous research suggests that cross-ownership can increase the influence of investors over investee firms. A significant body of research emphasizes the active participation of cross-owners in decision-making. For example, Kang et al. (2018) find that multiple blockholdings provide institutions with the necessary knowledge and experience to conduct effective monitoring. Similarly, He et al. (2019) highlight the role of institutional cross-owners in internalizing governance externalities by exercising voting rights. By contrast, Edmans et al. (2019) theorize that the alternative mechanism, the threat of exit, may also be reinforced under cross-ownership because being sold by cross-owners reveals the losers in the portfolio. In effect, the threat of exit is not generally accepted as a form of shareholder activism. According to Admati and Pfleiderer (2009), the impact of exit threats on corporate governance can vary greatly depending on the specific agency concerns of the firm. In the case of managerial shirking, the threat of exit may deter managers from maximizing firm value. Despite extensive research on the disciplinary effects of cross-ownership (Kang et al., 2018; He et al., 2019; Ramalingegowda et al., 2021), there are reasons to doubt that it aligns with managers' and shareholders' interests in all cases. Therefore, we investigate the effect of institutional cross-ownership on firm risk-taking in a transitional market.

We focus our attention on the particular context of institutional cross-ownership in China. Jiang and Kim (2015) suggest that compared with their US counterparts, institutional investors in China may face challenges in monitoring firms, mainly owing to their relatively low shareholding and lack of long-term perspective. However, these investors can still influence the market by selling shares,

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especially after the implementation of the split-share reform. Although it is difficult to be actively involved, institutional ownership is essential for a firm to improve its reputation among minority shareholders (Claessens and Fan, 2002). In China's A-share markets, which are primarily influenced by individual investors, the lack of transparency can lead to a situation where other traders imitate institutional behavior, thereby exerting pressure on prices (Tan et al., 2008). In short, findings from previous literature on the US may not be applicable to China.

In addition, a key differentiator of Chinese corporate governance is the weak legal system. Despite various protective measures enacted through legislation, China has been criticized for its poor legal environment. In the case of law enforcement, China received relatively lower scores than other countries. In the 2022 annual Rule of Law Index published by the World Justice Project, China ranked 95th out of 140 countries worldwide. When legal protections for investors are weak, managers may have greater latitude to act in their own interests. Therefore, special attention should be paid to corporate governance in transitional economies with weak legal enforcement.

Collectively, we have articulated two divergent views of how institutional cross-ownership may affect corporate risk-taking. The "communication" view suggests that institutional cross-ownership induces firms to pursue riskier but more valuable investments. It has been suggested that diversified owners tend to be less concerned about firm-specific risks because these risks can be easily spread out (Faccio et al., 2011; DesJardine et al., 2023). Getting involved in riskier projects is crucial not only for the value of the focal firm (Wright et al., 1996; Faccio et al., 2011; Koirala et al., 2020), but also for cross-owners to gain greater marginal benefits from peer firms (He et al., 2019). Thus, we estimate that institutional cross-owners are incentivized to guide managers toward higher levels of risk-taking. Moreover, institutional cross-owners can reduce monitoring costs by exploiting informational advantages through economies of scale (Ramalingegowda et al., 2021). Considering the potential benefits outweigh the costs of governance, it is expected that institutional cross-ownership will play an active monitoring role and engage firms in high-risk ventures.

On the contrary, the "signal" view contends that institutional cross-ownership strengthens the threat of exit and inadvertently reduces the level of corporate risk-taking. Under cross-ownership, being sold sends a stronger signal to the market about a company's poor performance. In the event of a liquidity shock, institutional cross-owners will sell the worst assets in their portfolios while retaining the good ones (Edmans et al., 2019). Since they have an information advantage over the public, other investors may mimic their trading decisions and exit the same securities (Bikhchandani et al., 1992; Xu et al., 2022). Given the potential negative consequences, managers may be unwilling to take any risk at first. Even if managers take productive actions, institutional investors will sell their shares if they observe that the increase in value does not meet their expectations (Admati and Pfleiderer, 2009). Then, a sale by an institutional cross-owner leads to a magnified decline in the share price and elicits ex-ante managerial reactions. When managers are party-contracted to market performance, they worry that their personal wealth and career success will be affected by the downward pressure (Dou et al., 2018; Döring et al., 2021). Therefore, managers tend to choose less risky actions in order to protect their own self-interest, as high-risk actions could potentially send negative signals to the market in the short-term. For the above reasons, institutional ownership may unintentionally reduce the degree of corporate risk-taking owing to the presence of exit threats.

To address this debate, we seek to examine the relationship between institutional cross-ownership and corporate risk-taking using a sample of Chinese listed firms over the period 2007–2019. Consistent with the "signal" view, our empirical tests suggest that institutional cross-ownership is negatively associated with corporate risk-taking. These findings withstand a series of robustness checks and continue to hold after addressing endogeneity concerns. We also conduct several additional analyses to strengthen our main point. We show that the effect of cross-ownership on corporate risk-taking is more pronounced when the exit threat is more credible or when managers are more concerned about market performance. The cross-sectional analysis is consistent with the view that capital market pressure serves as the channel through which institutional cross-ownership influences corporate governance. Furthermore, we find that the predictive power of institutional cross-ownership is attributable to active institutions that trade on short-term value. This is because active short-term institutional owners are more likely to exit when the valuation created in the short run is small. Finally, since previous studies have documented the positive sides of institutional cross-ownership, we expect that its threat of exit would produce positive and negative outcomes depending on agency problems. From a beneficial effect perspective, the results show that the possibility and degree of financial statement irregularities would be lower in response to the disciplining effect of the exit threat.

This study contributes to the literature in several ways. First, we add to the growing literature on the impact of institutional cross-ownership on corporate decision-making. A major strand of research has been devoted to explaining advances in corporate governance by highlighting the increased importance of cross-ownership in monitoring (e.g. Kang et al., 2018; Ramalingegowda et al., 2021). These studies argue that institutional cross-holders possess the ability to engage in efficient monitoring as a result of heightened capacity and incentives to participate. In contrast to these studies, we ask whether the effects of institutional cross-holders hold true for Chinese firms. Our results differ from those of previous studies, in that we find that the exit threat of institutional cross-ownership actually reduces managers' motivation to take additional risks to counteract downward price pressure. According to Xu et al. (2022), Chinese institutions prioritize short-term interests and often lack the patience to trade based on the long-term fundamental value of firms. We confirm that the exit threat is largely the result of active short-term investors.

Second, we shed light on the mechanism by which institutional investors profoundly influence corporate governance. Although cross-ownership has received extensive research attention, the significance of the institutional cross-ownership exit threat in corporate governance has been largely neglected. In particular, the theoretical model of Edmans et al. (2019) outlines that the threat of exit by cross-ownership will be reinforced as it signals to the market whether the assets are winners or losers in the portfolio. It has long been questioned whether institutions actively engage in management activities, especially in a transitional economy such as China (Jiang and Kim, 2020). According to McCahery et al. (2016), institutions ideally favor the use of the exit threat as an alternative mechanism in corporate governance. Based on our findings, it appears that institutional cross-ownership continues to rely primarily on the threat of exit as the main approach, rather than actively seeking engagement.

Third, this study contributes to existing knowledge about the factors that influence corporate risk-taking. Previous research has mainly examined how ownership structure influences the development of strategies for risky and uncertain projects (e.g. Boubaker et al., 2016). While institutional cross-ownership has been extensively studied in a variety of fields, the risk-taking behavior of managers has received less attention. We contend that within a Chinese setting, the existence of institutional cross-ownership has the potential to worsen conflicts of interest in corporate risk management, resulting in unfavorable consequences. Our study supports the theoretical analysis of Admati and Pfleiderer (2009), which suggests that the possibility of exit can discourage managers from taking value-maximizing actions and lead to dysfunctional investor behavior.

2. Related literature and hypothesis development

2.1. Determinants of corporate risk-taking

Given the importance of firms' risk-taking behavior, current research efforts are exploring aspects that may influence managers' attitudes toward risk. Based on the principles of agency theory, it can be argued that managers exhibit risk-averse behavior owing to their dependence on a single source of wealth and their concerns about career advancement (Wright et al., 1996; Kempf et al., 2009). As a result, prior research has placed considerable emphasis on mitigating managerial myopia and deterring agent shirking. Researchers initially focus on examining the incentive system that motivates managers to take more risks. For instance, Sanders and Hambrick (2007) demonstrate that when managers are provided with stock options, they are more likely to take high-variance gambles. Another area of academic research is the study of ownership structure as an important variable influencing corporate risk attitudes. Considering the monitoring activities of equity stakeholders, Faccio et al. (2011) claim that firms owned by diversified large shareholders are more likely to engage in risky investment activities than firms owned by non-diversified large shareholders. In contrast, the largest controlling shareholders tend to exhibit a preference for risk aversion owing to the limited availability of diversification options (Zhang, 1998). With the exception of the largest controlling shareholder, the monitoring function of other shareholders effectively limits expropriation actions and thus encourages a higher level of risk-taking (Boubaker et al., 2016). In sum, a firm's internal governance structure is likely to affect managers' interests and influence their risk-taking behavior.

In light of the above, we argue that few studies have connected corporate risk-taking with governance by introducing institutional cross-ownership, especially in the context of transitional economies. Thus, our study contributes to the literature by examining the consequences of cross-ownership structures in Chinese firms.

2.2. Governance through voice or exit

Large shareholders can exercise governance over firms through voice or exit (Shleifer and Vishny, 1986). The mechanism that has received the most academic attention is active monitoring through voice, which involves exercising voting rights and submitting proposals at shareholder meetings. Another mechanism that has received relatively little attention is the threat of exit, which can induce managers to react ex-ante even in the absence of actual action.

Despite the fact that institutional investors hold relatively large fractions of shares in their portfolio companies, their participation in corporate governance based on overt forms of monitoring or activism efforts is limited (Edmans et al., 2019). This might be explained as follows. First, large investors are reluctant to monitor management because of the free-rider problem. They have to bear the costs of monitoring and controlling management, while the benefits of their activism are shared by all shareholders (Shleifer and Vishny, 1986). Institutional investors may not capture all the benefits, because they typically hold relatively small stakes in firms. In contrast, other investors can capture the substantial value simply by selling their shares. Second, the legal and regulatory environment may discourage institutional active participation. For example, the diversification rule[1] directly affects the size of an institution's ownership stake, which is not large enough for effective control. The acting in concert rule[2] may discourage institutions from exercising power over corporate strategy because of the legal risk of misreporting. These rules place hurdles on how many shares institutions can own and what they can do with those shares. Third, without proper oversight and sufficient incentives, institutions may not actively use shareholder proxy rights (Bebchuk et al., 2017). Their exercise of voting rights is closely related to existing business relationships rather than concerns about the future of the company (CvijanoviĆ et al., 2016).

Meanwhile, the exit mechanism seems to be a viable alternative strategy for institutional investors. Large investors can exert influence through trading, which appears to be more feasible and less costly than voting. The results of a survey conducted by McCahery et al. (2016) show that institutional investors rely heavily on exits across all governance channels. This is consistent with the findings of Parrino et al. (2003), who present an empirical study that directly supports this view. However, in China, Jiang and Kim (2015) report that some institutional shareholders are considered small blockholders, despite their rapid growth rate. A natural question at this point is whether their exit threat is credible if they cannot engage. Researchers argue that blockholders are able to govern through the exit, regardless of the availability of voice. In the influential theoretical study by Edmans (2009), blockholes can depress the stock price by selling their shares, thereby incorporating private information into prices. This ex-post punishment scrutinizes managerial behavior

[1] The 2004 Fund Management Company Measures issued by the China Securities Regulatory Commission ("CSRC") published the diversification rule: any fund cannot hold more than 10% of its net assets in the shares of one listed company.

[2] According to the 2002 Shareholding Change Report Guideline issued by CSRC, shareholders shall disclose their purpose, formation date and plan once they form a concerted party through either a formal or informal agreement.

even when investors face disincentives to engage. In contrast, Admati and Pfleiderer (2009) support the notion that the threat of exit as an alternative voice mechanism generally mitigates agency problems based on private information. However, they question the effectiveness of exit threats in addressing any type of agency problem. Their theory predicts that the threat of exit reduces agency costs when the action induces managers to secure private benefits that are undesirable for shareholders ("bad action"). Meanwhile, when the action is costly for managers but desirable for shareholders ("good action"), the threat of exit can even exacerbate agency costs. More recently, Döring et al. (2021) provide empirical evidence that institutional shareholders regulate "bad action" agency problems more effectively through voice. Together, these studies highlight the importance of market pressure in corporate governance.

2.3. Institutional cross-ownership and corporate risk-taking

The "communication" view contends that institutional cross-ownership has a positive effect on corporate risk-taking by actively participating in corporate governance and reducing managers' risk aversion. This view contends that institutional cross-owners have stronger incentives to engage firms to take on more risk. In general, a well-diversified owner can diversify firm-specific risk and bear more risk than individuals (Faccio et al., 2011; DesJardine et al., 2023). In contrast to cross-owners, managers tend to be risk-averse because their compensation relies solely on the performance of the focal firm (Wright et al., 1996). Since theoretical and empirical studies have shown that varying uncertainty boosts the future value of a firm (e.g. Wright et al., 1996; Faccio et al., 2011; Koirala et al., 2020), cross-owners are more likely to require managers to undertake riskier projects. In particular, risk shifting by one firm can have similar effects on its peer firms. Bloom et al. (2013) strengthen the positive effect of R&D on gross social returns through spillovers. Further, Anton et al. (2021) theoretically demonstrate that firm innovation is sensitive to the internalization of knowledge externalities by cross-holders. Relative to holding a single firm, cross-holders may earn higher marginal returns for the same amount of monitoring inputs because they also receive benefits from all other peer firms (He et al., 2019). Thus, to maximize portfolio value, institutional investors should engage more actively with portfolio firms to pursue riskier investments.

Moreover, by holding more than one company in an industry, diversified owners have an informational advantage that can reduce monitoring costs. Through the communication channel, institutional overlap may even unintentionally grasp quantitative knowledge (Jung, 2013). For instance, firm managers may seek advice from institutional investors about how their peers respond to a newly accepted disclosure policy. In this case, cross-holders can gradually discover private information about a firm's behavior. Furthermore, cross-holding fund managers may share information during the coordination activities (Azar et al., 2018). Those industry-level commodities (e.g., similar strategy canvas, business environment, and disclosure policies) constitute the same considerable fixed costs involved in the costs of monitoring in the same industry (Ramalingegowda et al., 2021). The governance-relevant knowledge obtained from holding the portfolio ameliorates the process of obtaining and processing industry-specific information. Based on mixed integrated information, institutional investors can obtain a better understanding of the companies and identify more sources of investment.

According to these arguments, institutional cross-ownership has a positive effect on firms' risk-taking through increased incentives and monitoring capacity. Thus, we postulate our hypothesis as follows:

Hypothesis H1a. Institutional cross-ownership enhances corporate risk-taking.

On the contrary, the "signal" view suggests that institutional cross-ownership inadvertently lowers the level of corporate risk-taking through the threat of exit. The main effect of a diversified portfolio is that an institution's selling choices transmit more information into the stock price (Edmans et al., 2019). If an institution holds multiple stakes, a sale sends a stronger negative signal to the market, implying that the company being sold is the only bad one in the portfolio. As shown by Huang et al. (2023), mutual funds always sell losers to satisfy creditor demands, resulting in a persistent effect on security prices in an asymmetric information environment. Once cross-holders transmit signals to the market, the information cascade theory suggests that the observing investors are likely to follow the actions of their predecessors instead of relying on their private information (Bikhchandani et al., 1992). Such herding behavior leads to a low price ex-post and induces market performance concerns ex-ante, as in Edmans (2009) and Admati and Pfleiderer (2009). Therefore, multiple holders may have a stronger consequential influence on managerial actions through the threat of exit.

To avoid possible downward pressure, managers may be unwilling to take any risks at the beginning. Theoretical research predicts that if the public can observe what actions management takes, but only blockholders can observe the realized value of the company, then they will be inclined to sell their shares if they perceive that the value added is small (Admati and Pfleiderer, 2009). This is particularly true in China, where the investment logic of some large institutional shareholders is to seek immediate short-term gains rather than fundamental value (Xu et al., 2022). Downward pressure on prices reduces managers' efforts because their productive activities can easily send bad signals to the market in the short run. After all, short-term stock prices can directly affect managers' succession plans, professional reputation, and personal wealth. For instance, some stakeholders (e.g. credit-rating agencies) are less informed and can only learn from the information contained in the market (Bond et al., 2012). Since credit ratings are heavily influenced by price movements, a downward trend in prices may increase managers' concerns about a firm's ability to raise capital. Regarding short-term price effects, managers may reduce their risk-taking behavior because the riskier the actions they take, the more negative signals they may send to the market in the short term. According to this line of reasoning, the threat of exit may inadvertently exacerbate the agency problem and lead to lower levels of corporate risk-taking.

The key assumption underlying the above literature is the essential agency problem that the interests of managers and shareholders are not well aligned. As a developing country on the path to a market-oriented economy, China is characterized by a weak albeit improving legal environment (Jiang and Kim, 2015). In the absence of strong legal protections, managers have a certain latitude in making strategic choices that advance their own interests. Therefore, given the possible shock from cross-owners, managers are more

inclined to choose risk-averse strategies.

Based on the previous discussion, we argue that the exit threat is primarily used by institutional cross-ownership, which unintentionally induces insufficient risk-taking. Hence, we conjecture our hypothesis as follows:

Hypothesis H1b. Institutional cross-ownership suppresses corporate risk-taking.

3. Data and research design

3.1. Sample and data sources

This paper starts with a large sample of A-share companies listed on the Shanghai and Shenzhen stock exchanges from 2007 to 2019. Our sample starts in 2007 because China's split-share reform to eliminate non-tradable shares was completed by the end of 2007. Since the proxies for corporate risk-taking are based on a three-year rolling window, we obtain financial data from 2007 to 2021, and our sample period ends in 2019. We derive the detailed quarterly institutional ownership data from the CSMAR database. State ownership information is obtained from the CNRDS database. Other financial information is obtained from the CSMAR database. Following convention, we exclude financial listed firms, special treatment listed firms, and firms with missing variables. To reduce the impact of outliers, we winsorize all the continuous variables at 1% in each tail. The final sample consists of 25,888 firm-year observations associated with 3380 firms.

3.2. Major variable measures

3.2.1. Dependent variables

Following the literature (Boubakri et al., 2013; Langenmayr and Lester, 2018), we construct two measures of risk-taking based on the volatility of return on assets because riskier firms exhibit more income uncertainty. The first proxy variable (*Risk1*) is calculated as the standard deviation of *ROA* over three-year periods where *ROA* is defined as the ratio of earnings before interest and taxes (EBIT) to total assets. For instance, the level of corporate risk-taking in 2019 is measured by the volatility of *ROA* from 2019 to 2021. We use three-year windows as the basis for the calculation because the tenure of chief executive officers (CEOs) in China is generally three years. The second alternative proxy variable (*Risk2*) is the difference between the maximum and minimum *ROA* over a three-year interval. We multiply both measures by 100 for display in all the regressions.

3.2.2. Independent variables

For institutional cross-ownership, we follow He and Huang (2017)'s approach to construct the proxies for institutional cross-ownership at the firm-year level. We focus on institutional investors that own at least 5% of a firm's outstanding shares, since these blockholders exert significant influence on corporate governance. We define our primary measure, *CrossDum*, as a dummy variable that equals one if a firm is ever cross-owned by institutional investors in any quarter of a fiscal year, and zero otherwise. Thus, *CrossDum* captures the effect of institutional cross-ownership of a given firm. Our results are consistent with the following continuous proxies for institutional cross-ownership (untabulated). The second measure, *CrossNum*, is the number of unique institutions that cross-own the firm, averaged over the four quarters. The third one, *CrossShare*, is the aggregate share of all institutional cross-owners, averaged over the four quarters.

3.3. Methods

We use the following regression model to investigate the relationship between institutional cross-ownership and corporate risk-taking:

$$\mathrm{Risk}_{i,t} = \beta_0 + \beta_1 \mathrm{CrossDum}_{i,t} + \beta_2 \mathrm{CVs}_{i,t} + \sum\nolimits_j \mathrm{Industry}_j + \sum\nolimits_t \mathrm{Year}_t + \varepsilon_{i,t}, \tag{1}$$

where *Risk* is one of the two measures of corporate risk-taking. *CrossDum* is an indicator variable that equals one if a firm has institutional cross-ownership, and zero otherwise. *CVs* denote the control variables that may be significant for corporate risk-taking. In particular, we include firm size (*Size*) to account for the ability and willingness to make aggregate investments (Langenmayr and Lester, 2018; Koirala et al., 2020). Next, we control for firm investment opportunities by including sales growth (*Growth*) and market-to-book share (*MTB*) (Langenmayr and Lester, 2018). Later, we use firm age (*FirmAge*) to account for systematic variation in the level of risk-taking associated with a firm's life cycle. We also consider leverage (*Lev*) to account for firms' access to financing (Tran, 2019) and capital expenditure (*Cap*) to control for firms' contribution to investment (Boubakri et al., 2013). Similarly, as suggested by Langenmayr and Lester (2018), we then control for a firm's ability to sponsor risky projects by accounting for return on assets (*ROA*). To capture the effect of managerial entrenchment, we include CEO duality (*Dual*) and CEO share (*CEOShare*) (Ferris et al., 2017). Given that improved corporate governance may reduce managerial myopia, we control for board monitoring by including board independence (*Indep*) (Gopalan et al., 2021). Also, we consider the governance structure, which may affect the relationship between corporate risk-taking and institutional cross-ownership (Faccio et al., 2011; Boubaker et al., 2016). Therefore, we use equity balance (*Balance*), state ownership (*StateShare*), and institutional investor ownership (*InstShare*) as additional variables. Finally, we include Chinese regional GDP per capita (*GDP*) to capture the influence of country-level growth on managerial decisions (Boubakri et al., 2013). We

apply industry and year fixed effects and cluster standard errors at the firm level.

4. Main results

4.1. Summary statistics

Table 1, Panel A, reports the descriptive statistics of the main variables. The mean of *Risk1* (*Risk2*) is 3.204 (6.021), while the minimum and maximum values are 0.112 (0.217) and 21.150 (38.440), indicating large differences in the level of corporate risk-taking in our sample. The next three rows summarize the proxies for institutional cross-ownership. The mean of *CrossDum* is 0.136, indicating that 13.6% of the listed firms in our sample have institutional cross-ownership. This value is much lower than that of US firms (He and Huang, 2017), and one possible explanation is that mutual funds are prohibited by regulation from owning more than 10% of a listed firm's common stock. The summary statistics of the key variables are generally consistent with those in prior literature (Gao et al., 2019). Table 1, Panel B, presents the univariate comparisons. We divide our sample into groups based on whether they have institutional cross-ownership. Preliminary results show that, on average, firms without institutional cross-ownership are more willing to take risks than firms with institutional cross-ownership, supporting the "signal" view (Hypothesis H1b).

4.2. Baseline results

In this section, we examine the relationship between institutional cross-ownership and corporate risk-taking in the full sample. Table 2 reports the results of the baseline regression. The coefficients of independent variables are negative and significant at the 1% level, indicating that institutional cross-ownership is negatively associated with corporate risk-taking. Next, we examine the economic significance of *CrossDum* on *Risk1* and *Risk2*. In columns (1) and (2), the coefficients of *CrossDum* indicate that when a firm moves from no institutional cross-owner to at least one institutional cross-owner, then it shows a decrease of 0.30 (0.55) in *Risk1* (*Risk2*). Given that the mean of *Risk1* (*Risk2*) in cross-owned firms is 2.70 (5.08), the magnitude of this effect is economically significant. We also note that the correlations between institutional ownership (*InstShare*) and firm risk-taking are negative and significant at the 1% level. This result is consistent with the work of Fan and Fu (2020), who find that, contrary to mature market expectations, institutions have little monitoring ability and instead exert more selling pressure. Since we have already included the proxy for institutions (*InstShare*), the coefficients of *CrossDum* can be interpreted as the incremental effect of institutional cross-ownership. This result supports H1b that institutional cross-ownership leads to lower levels of corporate risk-taking.

4.3. Robustness checks

4.3.1. Alternative measures of corporate risk-taking

To test the robustness of our results, we begin by re-estimating the baseline model with alternative measures of corporate risk-taking. First, we use idiosyncratic risk (*Risk3*) as a market-based measure of firm risk. Idiosyncratic risk captures unexplained firm-specific risk that is independent of market movements. Following Langenmayr and Lester (2018), we calculate the standard deviation of the Fama-French three-factor regression residuals. Second, we consider financial distress, since higher risk-taking is associated with the probability of financial crisis. Following Ohlson (1980), we use the O-score (*Risk4*) to reflect the probability of bankruptcy. Third, we use total investment activity (*Risk5*) because firms will allocate more capital to long-term investments if they exhibit higher levels of risk-taking behavior. We define a firm's total investment as non-cash assets scaled by total assets (Chen and Wang, 2019). Consistent with our expectations, all tests show consistent results as we observe negative statistical coefficients for *CrossDum* in the first three columns of Table 3.

4.3.2. Alternative threshold for defining institutional cross-ownership

In the previous tests, we keep observations for which institutions' ownership exceeds 5%. It can be argued that the threshold may seem arbitrary because if institutions own more shares, they will have more incentives to engage, rather than to trade. In turn, we use an 8% threshold to include more institutions that own a larger fraction of shares. The results in columns (4) and (5) in Table 3 show that our results hold after using the alternative threshold.

4.3.3. Subsample period analysis

We use an alternative sample to confirm our conjecture. In 2011, the CSRC's revision of the fair-trading system prevented investment committees and directors from interfering with portfolio managers' investment activities within their mandates. With greater autonomy, the exit threat of institutional cross-ownership is greater, leading to less corporate risk-taking. As shown in the last two columns of Table 3, we repeat our baseline regression only for the subsample of firms after the revision. The absolute coefficient estimates are much higher than those for the full sample, indicating that groups with higher exit threats have lower risk-taking. Again, the results are consistent with our hypothesis that the predominant exit threat from institutional cross-ownership reduces firms' risk-taking.

4.3.4. Alternative explanation exclusion

We also consider an important alternative channel that explains how institutional cross-ownership can potentially decrease firm risk-taking from an anti-competitive perspective. According to Azar et al. (2018), cross-ownership can limit firms' motivation to

compete and sustain high-profit margins, as the benefits of excessive competitive aggressiveness by one firm may offset the escalating costs of other firms in the portfolio. Therefore, we predict that the reduction in firm risk-taking is due to weakened managerial incentives affected by controlled competition. Theory predicts that market product competition acts as a complement to managerial incentives, so firms in less competitive industries have weaker managerial incentives (Karuna, 2007). According to this argument, the effect of institutional cross-ownership shall be more pronounced in industries with high market share concentration because the small number of dominant shareholders can easily coordinate their strategies by reducing intense competition. Empirically, we construct Herfindahl-Hirschman Indices (*HHI*) of market concentration. We include *HHI* along with its interaction term with *CrossDum*. We find that the estimated coefficients of the interaction terms are positive and statistically insignificant. This suggests that our results are not driven by an anti-competitive explanation. For brevity, the results are not shown, but they are available from the authors upon request.

4.4. Endogeneity mitigation

4.4.1. Accounting for omitted variable bias

Admittedly, we cannot completely rule out the omitted variable bias even though we have included relevant confounders in our baseline model. In particular, this problem is attributed to unobservable critical covariates. For example, managerial ability is related to both institutional preference and corporate risk-taking. Then, a possible relationship between corporate risk-taking and institutional cross-ownership may be due to the lack of measurement of managerial ability.

To mitigate the potential threat of not including all possible confounders, we perform a bounding test to assess whether our estimate is robust to omitted variable bias (Oster, 2019). The bounding approach identifies omitted variable bias after accounting for not only movements in the coefficients of the relevant variables, but also changes in the coefficient of determination in the models. As suggested, we follow the procedure to compute the bounding value.

We obtain a bound estimate from the changes in the treatment effect after including the observable covariates. The analysis relies on two pieces of information. First, we set R^2_{max} to 1.3 times the value of R^2 derived from the specification with all observable variables. The R^2_{max} limit is implied by Oster (2019) using empirical research from top journals. Second, we define the degree of selection on observables and unobservables (δ) as one. Following the suggestion of Oster (2019), one can expect that the most important controls have been selected in the models, so it seems reasonable to assume that the observables are at least as important as the unobservables. Given the degree of selection from observables to unobservables, we estimate the treatment effect as $[\widetilde{\beta}, \beta^*(R_{max}, \delta)]$ when there is a downward bias.[3]

Specifically, we use the user-written Stata command *PSACALC* to execute the bound analysis. Accordingly, we obtain (−0.2993, −0.1963)/(−0.5527, −0.3603) as the bounded interval for the dependent variable *Risk1/Risk2*. Obviously, the corresponding identified set excludes zero. This may indicate that the effect of institutional cross-ownership has some causal effect on firms' risk-taking. Overall, the results suggest that our main conclusion is unlikely to be driven by omitted variables.

4.4.2. Excluding the selection effect of institutional cross-ownership

The change in corporate risk-taking observed in our study can be attributed to a selection effect. This occurs when institutional cross-holders, who have their own risk preferences, choose to invest in firms with lower levels of risk-taking. We first examine whether institutions change their portfolio selection in response to firms' risk-taking in the previous year. Consistent with Wu et al. (2023), we create a portfolio selection dummy (*Cross*) that takes the value of one for firms that are held by institutional cross-owners for the first time in the current year, and zero otherwise. We then regress *Cross* on lagged firm risk-taking, expecting a negative relationship if self-selection is present. In the first two columns of Table 4, the coefficients of lagged firm risk-taking are not statistically significant, indicating that institutional cross-owners do not hold stocks in response to firm risk-taking. Next, we examine whether institutional cross-ownership plays a role in identifying firms with the potential for consistent future performance. Specifically, following Dickinson (2011), we create a maturity dummy (*Maturity*) that equals one if the firm is considered mature, and zero otherwise. Since mature firms have already manifested stable performance, we expect the lagged maturity dummy (*L.Maturity*) to be positively related to *Cross* if institutional cross-ownership prefers a lower level of corporate risk-taking. As shown in column (3) of Table 4, the coefficient on *L. Maturity* is not statistically significant, suggesting that manifestly stable performance is not one of the selection criteria for institutional cross-ownership. Overall, our results are inconsistent with the self-selection effect.

4.4.3. Instrumental variable method

Further, we use an instrumental variable (IV) method to address the endogeneity issues related to reverse causality and omitted variables. Similar to previous studies (e.g. Appel et al., 2016; Kang et al., 2018), we exploit the nature of the China securities index (CSI) composition as an instrumental variable. Since CSI indexes are value-weighted, institutions can select firms by tracking these indexes. Adjustments to index inclusion create notable exogenous changes in the ownership structure of firms. Although index allocations are rebalanced based on firms' market capitalization, the selection criteria are determined by an arbitrary rule. Thus, changes in ownership structure should be exogenous after conditioning on market capitalization (Appel et al., 2016). In this setting, the CSI

[3] Specifically, β^* can be approximated by the expression: $\beta^* \approx \widetilde{\beta} - \delta\left[\dot{\beta} - \widetilde{\beta}\right]\frac{R^2_{max} - \widetilde{R}^2}{\widetilde{R}^2 - \dot{R}^2}$, where $\widetilde{\beta}$ and $\widetilde{R}^2$ ($\dot{\beta}$ and $\dot{R}^2$) are estimated using conditional (unconditional) regression of Eq. (1).

entrance satisfies the exclusion restriction because the index adjustment is relatively random. In addition, the index adjustment satisfies the relevance condition because index changes are associated with changes in an institution's portfolio. For example, when a firm enters the CSI, the probability of having passive owners decreases, while the probability of having active owners increases (Kang et al., 2018). Even if the target firms are not affected by the index adjustment, a change in the ownership structure is likely to occur when the co-owned peer firms are on the index list. Therefore, we use an IV approach based on the index listings. For investors, the CSI 300 and CSI 500 are popular benchmarks for tracking strategies. We create a CSI entrance dummy (*DCSI)* that takes the value of one if a stock switches from no index to either the CSI 300 or CSI 500, and zero otherwise.

We report the regression results from columns (4) to (6) in Table 4. In the first-stage regression estimation, the coefficient estimate of *DCSI* is negative and significant at the 1% level, indicating that the relevance condition holds. In the second-stage estimation, the coefficients of *CrossDum* are generally negative and significant. When we use an instrument for institutional cross-ownership, an increase in institutional cross-ownership leads to a decrease in corporate risk-taking. The IV estimation confirms the results of our baseline results.

5. Cross-sectional evidence

This section discusses the underlying mechanism in more detail. As discussed above, the mechanism underlying H1b is that the exit threat from institutional cross-ownership triggers managerial behavior. Therefore, we expect the negative effect of institutional cross-ownership to be more pronounced when the exit threat is more credible or when managers are more concerned about market performance. Specifically, we examine the role of three factors: stock liquidity, blockholder competition, and state ownership. Our cross-sectional evidence is consistent with the capital market pressure channel through which institutional investors influence corporate risk-taking.

5.1. Role of stock liquidity

By reducing the cost of exit upon weak earnings, liquidity allows dissatisfied institutional cross-owners to exit rather than intervene (McCahery et al., 2016). With increased liquidity, an institution can trade more aggressively to avoid disappointing firm performance (Dou et al., 2018). Conversely, market illiquidity prevents institutions from selling in the short term. Low liquidity forces institutions to bear the costs of intervening in firms and actively participating in corporate governance (Edmans et al., 2013). Thus, liquidity encourages investors to trade in large volumes and incorporate more information into stock prices. Because managers are sensitive to changes in stock prices, they are more likely to preserve private utility and therefore take less risk. Since liquidity increases the threat of exit, we expect the negative effect of institutional cross-ownership on corporate risk-taking to be more pronounced as liquidity increases.

We include the negative of Amihud's (2002) illiquidity ratio (*Liquidity*) and its interaction with *CrossDum* in our regressions. Table 5 reports the effect of liquidity on the relationship between institutional cross-ownership and corporate risk-taking. Consistent with the above augments, we find in columns (1) and (2) that the negative impact of institutional cross-ownership on corporate risk-taking is more pronounced as liquidity increases. The evidence suggests that with higher liquidity, institutional cross-ownership is more effective in exacerbating managerial short-termism and inducing lower levels of corporate risk-taking.

5.2. Role of blockholder competition

As suggested by Dou et al. (2018), competition among multiple blockholders increases the threat of exit through their aggressive trading. Large stakes may be necessary for investors to exercise governance, but not desirable if they want to exit troubled firms (Edmans, 2009). On the contrary, the cost of coordination among multiple blockholders hinders intervention and motivates them to compete for trading profits (Dou et al., 2018). Then, competition among blockholders will intensify the informativeness of the stock price, and thus block sales will lead to a less attractive price (Edmans and Manso, 2011). Overall, we expect blockholder competition to facilitate the threat of exit from institutional cross-ownership on managers' risk-taking behavior.

Accordingly, we measure blockholder competition(*BhComp*) using the Herfindahl index of blockholder ownership multiplied by −1 (Dou et al., 2018). To measure the impact of blockholder competition on the relationship between institutional cross-ownership and corporate risk-taking, we include the interaction term between *BhComp* and *CrossDum* in our model. As shown in columns (3) and (4) of Table 5, the coefficients of the interaction terms are negative and statistically significant, suggesting that the negative effect of institutional cross-ownership on managers' risk-taking behavior is stronger when blockholder competition is higher. The evidence is consistent with our prediction and strengthens our exit threat argument.

5.3. Role of state ownership

Finally, we examine the moderating effect of state ownership. Despite the transitional process of share privatization, state ownership is still prevalent in Chinese listed companies. Unlike private owners, the state is more likely to achieve socio-political goals to secure the political interests of bureaucrats. Apparently, wealth maximization is not the only objective for the state; rather, political self-interest may encourage politicians to engage in decision-making activities to realize a broader set of goals (Prabowo et al., 2018). Because the state is less concerned about short-term performance, its intervention provides managers with better protection against price declines to some extent. In particular, the state can reduce the tempting power of incentive-based compensation. Chinese SOE

managers are not incentivized by Western-style compensation, because they are not paid much more for better performance (Jiang and Kim, 2020). Since some managers are willing to return to government posts at the end of their tenure, they may be promoted and rewarded with non-monetary benefits later (Jiang and Kim, 2015). According to this view, we expect the effect of institutional cross-ownership to be more pronounced for firms with less state ownership if the exit threat is the underlying channel.

We measure state ownership as the proportion of shares owned by the government (*StateShare*). The indicator objectively denotes the degree of state ownership as the ownership structure rarely changes. The higher the proportion of state-owned shares, the lower the pressure from the capital market. We include both *StateShare* and *StateShare*×*CrossDum* in our main equation. Columns (5) and (6) in Table 5 show the positive association of the interaction term, which supports our conjecture. Overall, this evidence suggests that firms with fewer state shares respond more strongly to the exit threat of institutional cross-ownership by reducing corporate risk-taking.

6. Further analyses

6.1. Types of institutions: who is responsible for the predictive power?

6.1.1. Short- versus long-term institutions

We further investigate the extent to which the predictive power of institutional cross-ownership can be attributed to institutions with varying horizons. According to Yan and Zhang (2009), short-term opportunists trade frequently and prioritize short-term value for speculative purposes. They regularly evaluate the performance and industry ranking of portfolio companies and sell shares when the performance of portfolio companies declines. In contrast, long-term institutional investors who follow a value investing strategy take a more active interest in a company's operations and closely monitor management's investment decisions (Chen et al., 2007). Consequently, firms with a higher proportion of short-term institutional investors are more likely to face a significant exit threat. One might expect that short-term investors are primarily responsible for the negative effects of institutional cross-ownership.

To test this hypothesis, we follow the approach of Yan and Zhang (2009) and distinguish between short-term and long-term institutions based on their average turnover rate rankings over the past two years. After sorting institutional investors into three tertile portfolios, we classify an institution as a short-term/long-term institution if its average turnover rate ranks in the top/bottom tertile. Then, *STCrossShare/LTCrossShare* measures the ratio of outstanding shares held by short-term/long-term institutional cross-owners. We include *STCrossShare* and *LTCrossShare* separately to our baseline equation separately in Table 6.

We find strong evidence that the negative impact of institutional cross-ownership is mainly driven by short-term institutions. For example, in columns (1) and (2), when controlling for *STCrossShare*, the absolute value of the coefficient of *CrossShare* declines (from −0.783 to −0.321) and is no longer significant, while the coefficient of *STCrossShare* is −1.680 and significant at the 1% level. In contrast, when both *LTCrossShare* and *CrossShare* are included in the regression equation, the untabulated results show that the coefficient of *CrossShare* remains significantly negative, while the coefficient of *LTCrossShare* is positive and significant at the 1% level. These results suggest that the predictive ability of *CrossShare* is completely subsumed by *STCrossShare*. Interestingly, we find that *LTCrossShare* increases the level of corporate risk-taking, suggesting that long-term cross-owners effectively monitor the management and suppress managerial myopia.

6.1.2. Active versus passive institutions

Next, we split short-term institutions into active cross-owners and passive cross-owners. According to our logic, the primary underlying mechanism of institutional cross-ownership in managerial decision-making is the threat of exit. However, since passive investments typically track indexes, they are unlikely to exit owing to changes in corporate decision-making (McCahery et al., 2016; Helling et al., 2020). Therefore, we expect the negative impact of institutional cross-ownership to be driven mainly by active short-term institutions.

Following Kim et al. (2019), we categorize institutional owners into passive and active owners according to institution type. In particular, we classify banks, trusts, and insurance companies as passive owners and investment companies and investment advisors as active owners. We denote *STCrossShare(Passive)/STCrossShare(Active)* as the aggregate shares held by passive/active institutional cross-owners, respectively. In Table 6, columns (3) and (6) show the results with active and passive short-term institutions. When controlling for *STCrossShare(Passive)* and *STCrossShare(Active)*, the coefficients of overall institutional cross-ownership are no longer significant. In contrast, the coefficients of *STCrossShare(Passive)* are positive and insignificant, while the coefficients of *STCrossShare (Active)* are negative and significant at the 1% level. Consistent with our belief, passive investors have fewer incentives to use the exit threat. In all, we empirically show that the prohibiting effect of cross-ownership on risk-taking is primarily driven by active short-term institutions.

6.2. Do risk-taking behaviors affect future firm performance?

We also examine whether a firm's long-term growth is affected by its willingness to take greater risks. In particular, if managers are risk-averse, they will forgo risky but valuable investment projects and hold excessive cash (Tran, 2019). Given the key role of tangible or intangible assets in achieving competitive advantage, the future performance of the firm may be negatively affected. Therefore, it is essential to understand the economic consequences of institutional cross-ownership.

This paper uses *TobinQ*, a measure of firm value, and includes it as the dependent variable in model (1), while adding the cross product of cross-ownership and firm risk-taking (*CrossDum*×*Risk*) to investigate the impact of firm risk-taking on future firm value driven by cross-ownership. Since the impact of institutional cross-ownership on firm value may be lagged, the response variables are

tested using periods $t+1$ to $t+2$. Consistent with our conjecture, the future firm value is less responsive to corporate risk-taking in firms held by institutional cross-ownership. For example, in column (1) of Table 7, the estimated coefficient of the interaction term (*CrossDum*×*Risk1*) is negative and significant at the 5% level. This result is robust to regressions over the next associated columns, suggesting that cross-owners weaken the positive relationship between risk-taking behavior and future firm performance. This can be explained by the tendency of managers to "leave money on the table" rather than embark on riskier but value-creating projects under market pressure from cross-owners.

6.3. Disciplining effect of institutional cross-ownership

More broadly, the threat of exit due to institutional cross-ownership may discipline managers and reduce undesirable behavior from the blockholders' perspective (Admati and Pfleiderer, 2009). The existence of "bad action" agency problems, such as corporate wrongdoing, can be effectively mitigated by the credible threat of exit. As informed blockholders, institutional investors can observe the true value of managers' actions and exit in response to the potential unexpected outcomes. Therefore, managers will reduce the extraction of private actions and align their interests with blockholders (Dou et al., 2018). As a result, the amplified exit threat will reduce the need for managers to engage in undesirable activities. In this case, the governance power stemming from institutional cross-ownership reduces the incidence of corporate misconduct.

We use the firm's probability and level of financial irregularities as proxies for corporate misconduct. First, we define *VIO* as whether the firm commits financial reporting irregularities. *VIO* takes the value of one as if the firm violates the law, and zero otherwise. Second, we define *Degree* as the severity of penalties, which is divided into four levels based on severity (Li and Li, 2020). Higher values of *Degree* indicate more penalties. For each measure of firm misconduct, the coefficient of *CrossDum* in Table 7 is negative and significant, suggesting that firm misconduct decreases in response to the exit threat of institutional cross-ownership.

$$\mathsf{VIO}_{i,t}\left(\mathsf{Degree}_{i,t}\right) = \beta_0 + \beta_1 \mathsf{CrossDum}_{i,t} + \beta_2 \mathsf{CVs}_{i,t} + \sum\nolimits_j \mathsf{Industry}_j + \sum\nolimits_t \mathsf{Year}_t + \varepsilon_{i,t}, \tag{2}$$

However, a reduction in corporate misconduct may also be due to the monitoring effects of institutional cross-ownership. Thus, it is difficult to disentangle the true effects. We then conduct an additional test to examine whether the exit threat is the dominant channel through which institutional investors act. As suggested by Admati and Pfleiderer (2009), "bad action" agency problems are more likely to occur in cash-rich firms. Consistent with this argument, we predict that the disciplining effect of exit threats is more effective in situations prone to free cash flow abuse. We use *Cash* to denote the free cash flows standardized by total assets at the beginning of the year. We then modify model (3) to include *Cash* and the interaction between *Cash* and *CrossDum*. The results reported in Table 7 show that the coefficients of the interaction between *Cash* and *CrossDum* are negative and significant, suggesting that the reduction in corporate misconduct following the presence of institutional cross-ownership is more pronounced for cash-rich firms. These results confirm that the threat of exit is the dominant corporate governance practice of institutions.

7. Conclusion

We examine the relationship between institutional cross-ownership and corporate risk-taking. Theoretically, institutional cross-owners can influence firm behavior through either voting or trading (Edmans et al., 2019). However, institutional investors are less inclined to actively participate in China. Since the threat of exit can have a detrimental effect on firm-specific risks, we examine the effects of institutional cross-ownership in the specific context of China.

Our results differ from those of previous studies of the US capital market, in that we shed light on the negative effects of institutional cross-ownership in China. In particular, we show that this practice can significantly exacerbate managerial myopia and lead to reduced corporate risk-taking. By holding multiple companies, institutional investors will sell the losers, sending a strong signal to the market, and the resulting downward pressure exacerbates managers' risk-taking myopia. Our cross-sectional tests support our hypothesis that institutional cross-owners influence corporate risk-taking primarily through capital market pressure. Our findings provide a unique perspective on the governance of institutional cross-ownership, suggesting that in a transitional market, institutional cross-ownership may not always align with the interests of managers and shareholders. Rather, under the amplified exit threat, managers undertake fewer risks in response to the downward price pressure. However, since previous studies have documented the positive aspects of institutional cross-ownership, we do not classify its exit threat as a negative attribute. In addition, we emphasize the positive impact of disciplinary channels in reducing corporate misconduct. Thus, we contribute to the current literature that the exit threat of institutional ownership may have different outcomes depending on the agency problem.

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CRediT authorship contribution statement

Wenjing Yin: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing. **Weiping Li:** Methodology, Software, Writing – review & editing, Supervision, Project administration. **Yumiao Yu:** Writing –

review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

No potential conflict of interest was reported by the authors.

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Appendices

Table 1
Summary statistics and univariate comparison.

Panel A: Summary statistics

VarName	Obs	Mean	SD	Min	Median	Max
Risk1	25,888	3.204	4.090	0.112	1.711	21.150
Risk2	25,888	6.021	7.565	0.217	3.262	38.440
CrossDum	25,888	0.136	0.343	0.000	0.000	1.000
CrossNum	25,888	0.092	0.243	0.000	0.000	1.609
CrossShare	25,888	0.038	0.114	0.000	0.000	0.585
Size	25,888	15.567	0.947	13.742	15.460	18.417
Growth	25,888	0.189	0.428	−0.535	0.116	2.792
MTB	25,888	2.092	1.835	0.182	1.561	10.413
FirmAge	25,888	2.789	0.375	0.693	2.833	4.127
Lev	25,888	0.435	0.206	0.054	0.431	0.892
Cap	25,888	0.052	0.049	0.000	0.038	0.235
ROA	25,888	0.054	0.067	−0.323	0.052	0.331
Dual	25,888	0.253	0.435	0.000	0.000	1.000
CEOShare	25,888	0.045	0.106	0.000	0.000	0.502
Indep	25,888	0.373	0.055	0.091	0.333	0.800
Balance	25,888	0.697	0.601	0.024	0.526	2.770
StateShare	25,888	0.055	0.142	0.000	0.000	0.660
InstShare	25,888	0.449	0.241	0.004	0.472	0.907
GDP	25,888	10.996	0.530	8.972	11.058	12.009

Panel B: Comparison between firms with and without institutional cross-ownership

	Non-cross-held		Cross-held		Difference	
	Mean	Median	Mean	Median	Mean	Median
	(1)	(2)	(3)	(4)	(1)–(3)	(2)–(4)
Risk1	3.283	1.747	2.699	1.506	0.584***	0.241***
Risk2	6.169	3.329	5.084	2.853	1.085***	0.476***
Size	15.500	15.410	16.020	15.840	−0.520***	−0.430***
Growth	0.192	0.119	0.167	0.102	0.025***	0.017***
MTB	2.168	1.635	1.609	1.112	0.559***	0.523***
FirmAge	2.780	2.833	2.839	2.890	−0.059***	−0.057***
Lev	0.425	0.419	0.497	0.508	−0.072***	−0.089***
Cap	0.052	0.037	0.053	0.040	−0.001	−0.002**
ROA	0.054	0.052	0.054	0.050	0.000	0.003***
Dual	0.268	0.000	0.155	0.000	0.113***	0.000***
CEOShare	0.050	0.000	0.015	0.000	0.034***	0.000***
Indep	0.373	0.333	0.368	0.333	0.005***	0.000
Balance	0.691	0.515	0.732	0.593	−0.041***	−0.078***
StateShare	0.048	0.000	0.098	0.000	−0.050***	0.000***
InstShare	0.427	0.446	0.590	0.602	−0.163***	−0.156***

This table reports the descriptive statistics for the variables in our regressions. Panel A shows the number of observations, mean, standard deviation, minimum, median, and maximum values. The sample consists of 25,888 firm-year observations over the sample period 2007–2019. Panel B presents the differences in characteristics of firms with and without institutional cross-ownership. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 2
Institutional cross-ownership and corporate risk-taking.

	Risk1	*Risk2*
	(1)	(2)
CrossDum	−0.2993***	−0.5527***
	(−2.7958)	(−2.7639)
Size	0.1967***	0.3584***
	(3.8290)	(3.7494)
Growth	0.1545**	0.2582*
	(2.0332)	(1.8491)
MTB	0.4159***	0.7773***
	(14.5693)	(14.7358)
FirmAge	0.1323	0.2557
	(1.0668)	(1.1049)
Lev	1.1535***	2.1988***
	(4.4512)	(4.5735)
Cap	−2.9611***	−5.2703***
	(−4.2629)	(−4.0629)
ROA	−19.3689***	−35.5484***
	(−23.1461)	(−23.0245)
Dual	0.2803***	0.5192***
	(2.8439)	(2.8437)
CEOShare	−1.3901***	−2.5592***
	(−3.0896)	(−3.0614)
Indep	−0.7120	−1.3124
	(−1.0377)	(−1.0304)
Balance	0.2603***	0.4727***
	(3.6276)	(3.5463)
StateShare	−0.8451***	−1.5731***
	(−3.9555)	(−3.9450)
InstShare	−1.2302***	−2.2855***
	(−5.8586)	(−5.8616)
GDP	−0.1678*	−0.3182*
	(−1.7149)	(−1.7466)
Industry & Year FE	Yes	Yes
Observations	25,888	25,888
Adjusted R^2	0.1624	0.1615

This table presents the impact of institutional cross-ownership on corporate risk-taking. *Risk1* is the standard deviation of *ROA* over three-year overlapping periods, while *Risk2* is the difference between the maximum and minimum *ROA* over a three-year interval. *CrossDum* is an indicator variable that equals one if an enterprise is ever cross-held by institutional investors in any quarter in a fiscal year, and zero otherwise. All regressions control for industry and year fixed effects. The t-statistics in parentheses are based on robust standard errors clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 3
Robustness checks.

	Alternative measures of risks			Alternative ownership proportions		Sub-sample period analysis	
	Risk3	*Risk4*	*Risk5*	*Risk1*	*Risk2*	*Risk1*	*Risk2*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CrossDum	−0.0005***	−0.3112***	−0.0215***	−0.4046***	−0.7561***	−0.3539***	−0.6555***
	(−3.3411)	(−2.8339)	(−5.4102)	(−3.4675)	(−3.4777)	(−2.8657)	(−2.8507)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry & Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,549	25,888	11,881	25,888	25,888	19,805	19,805
Adjusted R^2	0.4207	0.6283	0.3275	0.1627	0.1618	0.1775	0.1773

This table presents the results of robustness checks. The first three columns present the results of regressions of alternative measures for corporate risk-taking. *Risk3* is an idiosyncratic risk, measured as the standard deviation of the Fama-French three-factor regression residuals. *Risk4* is the O-score, which is the probability of bankruptcy. *Risk5* is a firm's total investment as firms will devote more capital to long-term investments if they exhibit higher levels of risk-seeking behaviors. The next two columns present the results of regressions employing CrossDum with an alternative threshold. The last two columns re-estimate the baseline regression for the year after the fair-trading system revision by the CSRC. All regressions control for industry and year fixed effects. The t-statistics in parentheses are based on robust standard errors clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4
Endogeneity mitigation.

	Exclude the selection effect of institutional cross-ownership			An instrumental variable method		
	Cross	*Cross*	*Cross*	*CrossDum*	*Risk1*	*Risk2*
	(1)	(2)	(3)	(4)	(5)	(6)
L.Risk1	−0.0003					
	(−1.2890)					
L.Risk2		−0.0002				
		(−1.3849)				
L.Maturity			−0.0004			
			(−0.2387)			
DCSI				−0.0352***		
				(−3.4141)		
CrossDum					−9.2143**	−17.2433**
					(−2.3772)	(−2.3799)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,648	21,648	21,648	25,888	25,888	25,888
Adjusted R^2	0.0119	0.0119	0.0119	0.0883	–	–

This table presents the results of endogeneity mitigation. The first three columns show regression results that exclude the selection effect of institutional cross-ownership. *Cross* takes the value of one for firms held by institutional cross-owners for the first time in the current year, and zero otherwise. *Maturity* equals one if the firm is considered mature, and zero otherwise. The next three columns employ an instrumental variable estimation. *DCSI* takes the value of one if a stock switches from no index to either CSI300 or 500 indexes, and zero otherwise. Column (4) presents the result of the first-stage regression while columns (5) and (6) present the results of the second-stage regression. All regressions control for industry and year fixed effects. The t-statistics in parentheses are based on robust standard errors clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 5
Cross-sectional evidence.

	Stock liquidity		Blockholder competition		State ownership	
	Risk1	*Risk2*	*Risk1*	*Risk2*	*Risk1*	*Risk2*
	(1)	(2)	(3)	(4)	(5)	(6)
CrossDum	−0.3903***	−0.7272***	−0.8512***	−1.5730***	−0.3870***	−0.7173***
	(−3.2996)	(−3.2949)	(−3.0764)	(−3.0484)	(−3.2769)	(−3.2584)
CrossDum×Liquidity	−0.1703**	−0.3253**				
	(−2.4091)	(−2.4393)				
CrossDum×BhComp			−0.8096**	−1.4968**		
			(−2.0664)	(−2.0497)		
CrossDum×StateShare					1.0071**	1.8887**
					(2.4022)	(2.4101)
Liquidity	0.1478***	0.2698***				
	(4.5063)	(4.4562)				
BhComp			0.7022***	1.2959***		
			(2.8137)	(2.7969)		
StateShare	−0.8568***	−1.5956***	−0.8459***	−1.5745***	−1.0823***	−2.0179***
	(−4.0064)	(−3.9978)	(−3.9736)	(−3.9623)	(−4.6321)	(−4.6273)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,888	25,888	25,888	25,888	25,888	25,888
Adjusted R^2	0.1631	0.1622	0.1631	0.1622	0.1626	0.1617

This table presents the results of cross-sectional evidence. *Liquidity* is measured as the negative of Amihud's (2002) illiquidity ratio. *BhComp* is measured for blockholder competition by using blockholder concentration. *StateShare* is the fraction of stakes owned by the government. All regressions control for industry and year fixed effects. The t-statistics in parentheses are based on robust standard errors clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6
Types of institutions: Who is responsible for the predictive power?

	Risk1			*Risk2*		
	(1)	(2)	(3)	(4)	(5)	(6)
CrossShare	−0.7830***	−0.3210	−0.3206	−1.4522**	−0.5926	−0.5919
	(−2.5844)	(−0.9724)	(−0.9711)	(−2.5683)	(−0.9654)	(−0.9641)
STCrossShare		−1.6795***			−3.1252***	
		(−2.9324)			(−2.9414)	
STCrossShare(Active)			−1.6917***			−3.1476***

(*continued on next page*)

Table 6 (*continued*)

	Risk1			*Risk2*		
	(1)	(2)	(3)	(4)	(5)	(6)
			(−2.9545)			(−2.9631)
STCrossShare(Passive)			1.9573			3.5662
			(0.2051)			(0.2027)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,888	25,888	25,888	25,888	25,888	25,888
Adjusted R^2	0.1623	0.1626	0.1626	0.1614	0.1617	0.1617

This table presents the analysis of different types of institutions. *CrossShare* is the aggregate shares held by all institutional cross-holders, averaged over the four quarters. *STCrossShare* measures the ratio of outstanding shares held by short-term institutional cross-owners. *STCrossShare(Active)* and *STCrossShare(Passive)* are denoted for the ratio of outstanding shares held by active and passive short-term institutional cross-owners, respectively. All regressions control for industry and year fixed effects. The t-statistics in parentheses are based on robust standard errors clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 7
Economic consequence analysis.

	Future firm value				The disciplining impact			
	F.TobinQ		*F2.TobinQ*		*VIO*	*DEGREE*	*VIO*	*DEGREE*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CrossDum	0.0864***	0.0875***	0.0826**	0.0850**	−0.4468***	−0.4419***	−0.6306***	−0.6266***
	(2.6852)	(2.7048)	(2.1040)	(2.1559)	(−2.8045)	(−2.7514)	(−3.4086)	(−3.4023)
CrossDum×Risk1	−0.0149**		−0.0157*					
	(−2.3015)		(−1.6749)					
CrossDum×Risk2		−0.0081**		−0.0088*				
		(−2.3198)		(−1.7527)				
CrossDum×Cash							−0.6382**	−0.6427**
							(−2.1971)	(−2.1934)
Risk1	0.0220***		0.0287***					
	(8.3771)		(7.5966)					
Risk2		0.0119***		0.0157***				
		(8.4776)		(7.6937)				
Cash							−0.1182	−0.1272
							(−0.9785)	(−1.0443)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry & Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	24,865	24,865	21,835	21,835	25,888	25,888	25,888	25,888
Adjusted R^2/Pseudo R^2	0.5530	0.5531	0.4430	0.4431	0.0554	0.0486	0.0561	0.0492

This table presents the analysis of economic consequences. *TobinQ* denotes firm value. VIO takes the value of one as if the firm violates legislation, and zero otherwise. *Degree* is the penalty severity divided into four levels based on severity. We use *Cash* to denote the free cash flows standardized by total assets at the beginning year. All regressions control for industry and year fixed effects. The t-statistics in parentheses are based on robust standard errors clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

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